Session 8 Assignment Problems - Introduction to Functions

1. Allow the user to repeatedly enter a quantity and price. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute the total (quantity times price). The function should be passed the quantity and price and then return the total. In the function, provide a 10% discount if the total is over $10,0000.00. Display quantity, price and total. Sum and display the extended price.

Input	Process	Output
Quantitiy	Function pass for quantity and price	Quantity
Price	Total (Formula) = Quan * Price	Price
Y/N	Total > 10,000,000 = 10 %	Total
	Answer == yes to continue	Sum
	ExtPrice = ExtPrice + Total	Extended Price

2. Enter players last name, number of hits and at bats at the keyboard. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute batting average. Pass the hits and at bats to the function. The function should return batting average. Display last name and batting average. Give a count of the number of players entered.

Input	Process	Output
lname	Pass hits and bats to function	lname
# of hits	Function RETURN bat avg	Bat avg
# of bats	Bat avg = hits / at bats	# of players total
Y/N	Answer == yes to continue	
	Keep running count of players	

3. Enter the destination city, miles travelled and gallons used for a trip. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute miles per gallon. Pass miles travelled and gallons used to the function. The function should return miles per

gallon. Count the number of entries made (number of trips) Display destination city, miles and mpg. At end display the number of entries made.

Inpuit	Process	Output
City Destination	Pass miles travelled and Gal used to function	City
Miles	Mpg = Miles / Gal used	Miles
Gal used	Function Return Mpg	Mpg
Y/N	Count entries / trips	# of entries

4. Allow the employee to enter last name, job code and hours worked. Prompt the user on whether they want to do the program (Yes or No). Use a function to determine the pay rate. Pass to this function the job code and it should return rate of pay. Use Job code L is $25/hr, A is $30/hr and J is $50/hr for respective pay rates. Compute gross pay. Give time and a half for overtime. Display last name and gross pay. Sum and display total of all gross pay.

Input	Process	Output
Iname	Function determine pay rate Pass function to jobcode	Lname
job code	Pay rate based on job code: L is $25/hr, A is $30/hr and J is $50/hr	Gross pay
Hrs worked	Gross Pay = hrs * pay rate	Total gross pay
Y/N	Time and ½ for ot	
	Funciton Return rate of pay	

5. Allow the user to enter student last name, credit hours and district code. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute tuition owed. Charge In district (code of I) $250 per credit hour. Out of district (code of O) is $550 per credit

hour. The function should receive credit hours and district code and return tuition owed. Display student name and tuition owed. Sum and display total of all tuition owed.

Input	Process	Output
Iname student	Function compute tuition owed	Iname
Credit hrs	Tuition rate based on district #	Tuition owed
District code	Compute tuition owed: Credit hrs * District I/O (250/550)	Total tuition owed
Y/N	Function return tuition owed	

01/07/2026